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UPES UNIVERSITY OF TOMORROW

Elements of AIML

Unit 02 – Lecture 02: Predicate Logic (First-Order Logic)

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Topic focus: **objects, relations, and quantifiers** for richer knowledge bases

Repository: `https://github.com/tali7c/Elements-of-AIML`

Sources: Russell and Norvig, *Artificial Intelligence: A Modern Approach* (4e), Ch. 7–9 (Logical Agents, First-Order Logic, Inference).;Huth and Ryan, *Logic in Computer Science* (2e), selected sections on propositional and first-order logic.;Genesereth and Nilsson, *Logical Foundations of Artificial Intelligence*, selected chapters.

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Why predicate logic?

Syntax

Semantics

Translation practice

Summary

Learning Outcomes

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- Interpret the meaning of $\forall$ and $\exists$ and understand scope.
- Translate simple English statements into FOL (and spot common pitfalls).

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- $\neg, \wedge, \vee, \rightarrow$: NOT, AND, OR, IMPLIES
- $A(x)$: predicate A applied to object x

Why Propositional Logic Becomes Verbose

Propositional logic treats facts as atomic symbols.

Example

"Ali is a student" and "Sara is a student" become two unrelated atoms:

$$Student(Ali),\ Student(Sara) \quad \text{(FOL)} \quad \text{vs} \quad p,\ q \quad \text{(PL)}$$

With many objects, PL needs too many symbols and cannot express general rules compactly.

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- **Quantifiers**: $\forall$ (for all), $\exists$ (there exists)

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- Build complex sentences using propositional connectives:

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- Add quantifiers:

$$\forall x \ Student(x) \rightarrow WorksHard(x)$$

Free vs Bound Variables (Scope)

- A variable is **bound** if it is under a quantifier.

Examples:

$$\forall x\ Likes(x, AI) \quad (x \text{ bound})$$

$$Likes(x, AI) \quad (x \text{ free; meaning depends on context})$$

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- A variable is **free** if it is not bound by any quantifier.

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Quantifiers (Meaning + Scope)

- $\forall x\ P(x)$: for every object x in the domain, $P(x)$ is true.

Common confusion

$\forall x\ \exists y\ Likes(x, y)$ is **not** the same as $\exists y\ \forall x\ Likes(x, y)$.

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- **Scope matters**: where the quantifier applies.

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Negating Quantifiers (Very Useful Later)

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These rules are key in clause form conversion (Lecture 03).

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Semantics (What Makes a Sentence True?)

To interpret FOL, we specify:

- a **domain** of objects (what x can be),

Model idea

A model is an interpretation where a sentence evaluates to **true**.

Semantics (What Makes a Sentence True?)

To interpret FOL, we specify:

- a **domain** of objects (what x can be),
- an **interpretation** that assigns meaning to constants, functions, and predicates.

Model idea

A model is an interpretation where a sentence evaluates to **true**.

A Tiny Semantics Example

Domain: $\{Ali, Sara\}$ Predicate: $Student(\cdot)$ Suppose interpretation says:

$$Student(Ali) = T, \quad Student(Sara) = F$$

Then:

$$\exists x\ Student(x) \text{ is true, but } \forall x\ Student(x) \text{ is false.}$$

Translation Patterns (English → FOL)

- "All A are B":

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$$\exists x\ A(x) \wedge B(x)$$

- "No A is B":

$$\forall x\ A(x) \rightarrow \neg B(x)$$

More Patterns: "Only", "Exactly One" (Optional)

- "Only A are B" means: if B then A

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- "Exactly one A exists" (sketch):

$$\exists x\ A(x)\ \wedge\ \forall y\ (A(y) \rightarrow y = x)$$

Quantifier Order Example

Statement 1

Every student is enrolled in *some* course:

$$\forall x\ Student(x) \rightarrow \exists y\ (Course(y) \wedge Enrolled(x, y))$$

Statement 2

There exists a single course that *all* students are enrolled in:

$$\exists y\ Course(y) \wedge \forall x\ (Student(x) \rightarrow Enrolled(x, y))$$

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- Forgetting to specify the domain (what objects exist?)
- Using $\forall x\ A(x) \wedge B(x)$ when you meant implication
- Wrong quantifier order (changes meaning)
- Ambiguous English ("someone" / "the" / "only")

Bridge to Next Lecture: Substitution & Unification (Pr

In resolution for FOL, literals may "match" after substituting variables. Example:

$$Likes(x, AI) \text{ matches } Likes(Ali, AI) \text{ with substitution } \{x/Ali\}.$$

We will formalize this as **unification** in Lecture 03.

Summary

- Predicate logic adds **objects and relations** to propositional logic.
- Quantifiers $\forall$ and $\exists$ express general rules compactly.
- Semantics requires a domain and interpretation.
- Next: clause form, resolution, and unification (Lecture 03).

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Exit Questions

1. Why is propositional logic insufficient for many knowledge bases?
2. Translate: "All humans are mortal" and "Some student likes AI" into FOL.
3. Explain why $\forall x \exists y\, P(x, y)$ differs from $\exists y \forall x\, P(x, y)$.
4. Give one example of an English sentence that is ambiguous without clarifying assumptions.